AUTOMATED PARKING SYSTEM USING MACHINE LEARNING

A PROJECT REPORT

Submitted to



Visvesvaraya Technological University

BELAGAVI - 590 018

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in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

In

Computer Science & Engineering

SMVITM

Department of Computer Science and Engineering

SHRI MADHWA VADIRAJA INSTITUTE OF TECHNOLOGY AND MANAGEMENT

Vishwothama Nagar, BANTAKAL – 574 115, Udupi District

2023 - 2024

SHRI MADHWA VADIRAJA INSTITUTE OF TECHNOLOGY AND MANAGEMENT

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ACKNOWLEDGEMENT

It is our privilege to express sincerest regards to our project guide Mr. Raghavendra Hegde, Asst.Prof.(Sr) Dept. of CSE, SMVITM, Bantakal for helping us in successful completion of this project work.

We would also like to thank our project coordinators Dr. Sadananda L, Associate Prof. Dept. of CSE, SMVITM, Bantakal for helping us in successful completion of this project work.

We would like to thank Prof. Dr. Soumya J Bhat, HOD Dept. of Computer Science & Engineering for his inspiration during the completion of the project.

We would like to express our gratitude to Prof. Dr. Thirumaleshwara Bhat, Principal, SMVITM, Bantakal for extending his support.

We take this opportunity to express our deepest gratitude and appreciation to all those who helped us directly or indirectly towards the successful completion of this project.

We would like to thank our teaching and non-teaching staff, friends, who supported and encouraged us.

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ABSTRACT

The Automated Parking System (APS) using Machine Learning represents a groundbreaking solution aimed at revolutionizing conventional parking paradigms through the application of advanced technologies. This innovative project focuses on the implementation of machine learning algorithms to streamline and automate various aspects of the parking process. Central to its objectives is the efficient detection and recognition of vehicles entering the parking facility, employing computer vision and deep learning for accurate identification, including license plate recognition. The system incorporates real-time monitoring of parking space occupancy, integrating sensors and cameras to provide up-to-date information on available spaces. A user-friendly interface or mobile application is developed to relay this information to users, guiding them seamlessly to the nearest available parking spot. Machine learning algorithms are employed to analyse data and predict parking space availability, optimizing space utilization. The overarching goal is to minimize the time and effort required for users to locate suitable parking spaces, contributing to improved traffic flow, reduced congestion, and an overall enhanced parking experience.

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Chapter 1

INTRODUCTION

1.1 Background

As urban areas continue to grow and more people acquire cars, the quest for parking spots becomes increasingly challenging. This surge in vehicle ownership exacerbates existing problems like traffic congestion, inefficient use of parking space, and heightened environmental pollution due to increased emissions. The traditional approach to managing parking, reliant on manual processes and lacking integration with modern technology, struggles to keep pace with these mounting issues. Such systems often employ outdated methods for allocating parking spaces, resulting in inefficiencies, long wait times, and frustration among drivers.

Moreover, traditional parking systems face inherent limitations in adapting to fluctuating demand and providing real-time updates to users. Drivers frequently encounter difficulties in locating available parking spots, leading to heightened vehicle emissions, increased fuel consumption, and overall urban congestion. Additionally, the absence of dynamic pricing mechanisms and effective enforcement tools exacerbates the challenges confronted by parking operators and urban planners in efficiently managing parking resources.

However, there is hope on the horizon with the advent of innovative technology solutions such as automated parking systems. These systems leverage a sophisticated array of technologies, including sensors, cameras, and advanced algorithms, to revolutionize the efficiency and efficacy of parking operations. By detecting available parking spaces, guiding vehicles to vacant spots, optimizing space usage through dynamic allocation, and forecasting future parking needs based on historical data and real-time analytics, automated parking systems offer a beacon of hope in addressing the pressing challenges of urban parking management. Moreover, the integration of machine learning allows these systems to continuously learn and adapt to changing conditions, thereby enhancing accuracy and responsiveness over time. As a result, automated parking systems not only alleviate congestion and optimize space utilization but also improve the overall urban mobility experience for residents and visitors alike.

1.2 Automated Parking System

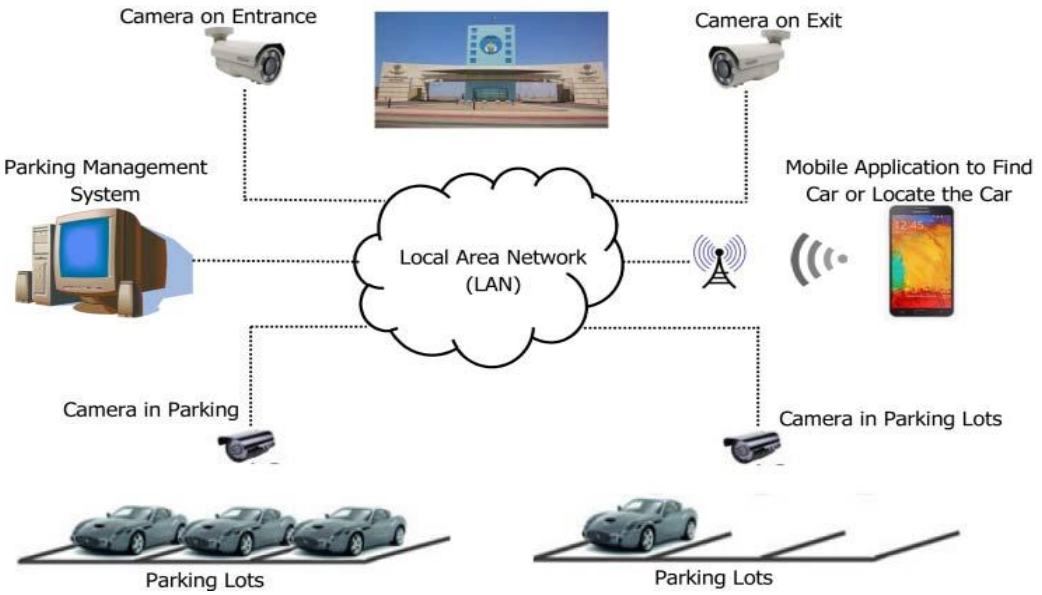


Figure 1.1: Automated Parking System

An automated parking system using machine learning, illustrated in Fig. 1.1, integrates sensors, cameras, edge devices, a central server, and a user interface to improve parking management. Sensors and cameras collect real-time data on parking spot occupancy, which is processed by edge devices running object detection models. This data is transmitted to a central server for advanced analysis and storage. The server uses machine learning models to optimize space allocation and predict future parking availability. Users receive real-time updates and guidance via a mobile app or web interface, enhancing efficiency and convenience. This system continuously adapts and improves based on new data, ensuring efficient parking space utilization.

1.3 Machine Learning

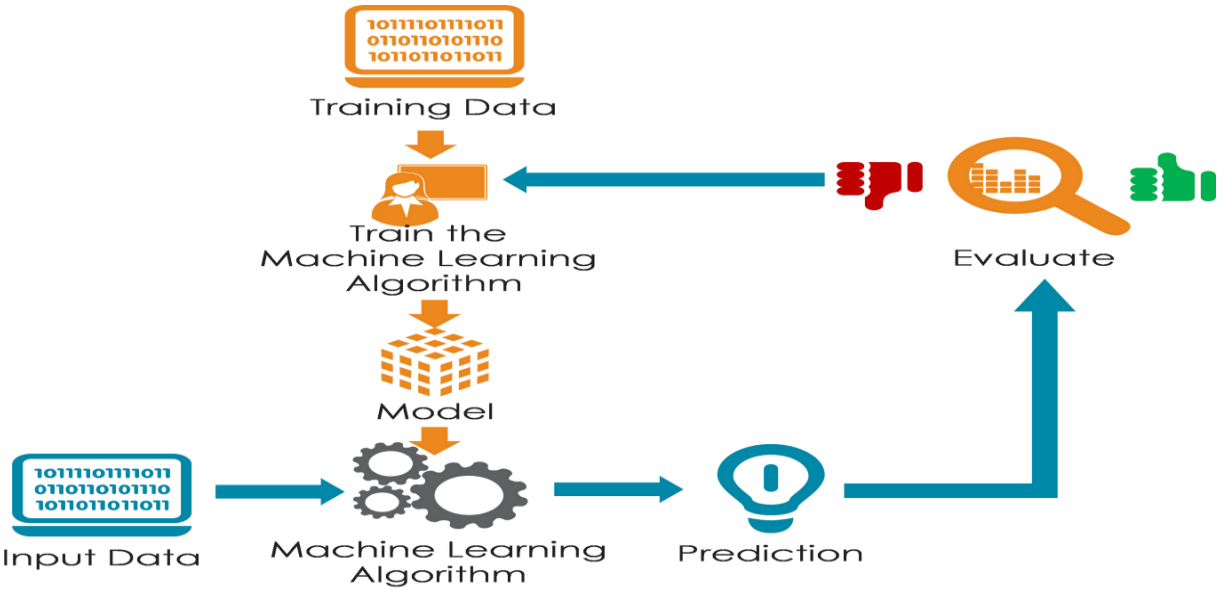


Figure 1.2: Machine Learning Model

Machine learning is a branch of artificial intelligence that empowers computers to learn and improve from experience without explicit programming. It encompasses a series of steps, starting with data collection from diverse sources, followed by preprocessing to clean and format the data for analysis. Then, a suitable machine learning algorithm is selected based on the problem and data type, and the model is trained using the collected data to recognize patterns and relationships. Subsequently, the model's performance is evaluated using separate testing data, and adjustments are made to optimize its parameters. Once trained and fine-tuned, the model is deployed into production to make predictions or decisions on new data. With its ability to analyze large datasets and extract meaningful insights, machine learning has found applications in numerous fields, from healthcare and finance to marketing and autonomous systems, revolutionizing industries and driving innovation.

1.4 Scope and Significance

The scope of this project lies in the development and implementation of an Automated Parking System (APS) specifically designed for commercial parking areas. The APS will harness cutting-edge technologies such as machine learning, image processing, and real-time data analysis to revolutionize parking management. Key functionalities include the accurate detection and recognition of vehicles entering and exiting the parking area, extraction of license plate information, real-time updates on available parking spaces, and seamless guidance for users to locate vacant spots efficiently. Additionally, the APS will integrate features for revenue collection, parking monitoring, and performance evaluation, ensuring comprehensive and efficient parking management solutions.

The significance of this project extends beyond mere convenience, as it addresses pressing urban challenges such as traffic congestion, environmental pollution, and inefficient space utilization. By optimizing parking space usage and reducing congestion, the APS contributes to smoother traffic flow, reduced emissions, and improved air quality in urban environments. Moreover, the system's user-friendly interface and real-time updates enhance accessibility and convenience for drivers, fostering a more inclusive and equitable urban mobility landscape. Furthermore, the APS has the potential to bolster economic productivity by increasing revenue streams for parking operators and businesses. Ultimately, this project represents a pivotal step towards smarter, more sustainable urban development, aligning with broader societal goals of enhancing quality of life and promoting innovation in transportation systems.

Chapter 2

LITERATURE REVIEW

In 2020, John Smith et al. proposed an automated parking system using machine learning techniques to enhance parking efficiency and user experience. The system employs a combination of computer vision and neural networks to identify available parking spots in real-time. Users can interact with the system through a mobile application, which provides a live feed of parking availability. The mobile app uses GPS to guide users to the nearest available parking space. Additionally, the system utilizes historical data and machine learning algorithms to predict parking spot availability at different times of the day, improving the accuracy of the recommendations provided to users. The system aims to reduce the time spent searching for parking and to optimize the use of available parking spaces.

In 2018, Sara Johnson et al. proposed a smart parking system that integrates IoT and machine learning to monitor and manage parking spaces. The system uses sensors to detect the presence of vehicles and a machine learning algorithm to analyse occupancy patterns. The data collected is processed in real-time and displayed on a mobile application, which helps drivers find available spots quickly. The system also includes a predictive feature that anticipates parking availability based on historical data and current trends.

In 2019, Michael Lee et al. introduced a deep learning-based parking management system that utilizes convolutional neural networks (CNNs) to analyse images from parking lot cameras. The system identifies empty and occupied parking spaces with high accuracy. Drivers can access this information through a mobile app, which provides real-time updates and navigational assistance. Additionally, the system can forecast future parking space availability, helping users plan their trips better.

In 2021, Priya Singh et al. developed a machine learning-driven parking system that uses a combination of reinforcement learning and computer vision. The system employs cameras to monitor parking lots and a reinforcement learning algorithm to optimize parking spot allocation

dynamically. The mobile application linked to the system offers real-time parking information and predictive analytics to users, aiming to minimize the time and effort spent searching for parking.

In 2022, Alex Kim et al. presented an intelligent parking system that leverages big data analytics and machine learning. The system collects data from various sources, including sensors, cameras, and user inputs, to analyse parking usage patterns. Machine learning algorithms are used to predict parking demand and optimize space allocation. The system features a user-friendly mobile app that provides real-time information on available parking spots and routes users to them efficiently.

In 2023, Emily Davis et al. designed an automated parking solution that combines machine learning with edge computing. The system uses edge devices installed in parking lots to process data locally, reducing latency and improving response times. Machine learning models analyze data from cameras and sensors to detect parking availability and predict future occupancy. The mobile application connected to the system offers real-time updates and predictive insights, enhancing user convenience.

In 2024, David Martinez et al. introduced a hybrid approach to automated parking systems using both machine learning and blockchain technology. The system employs machine learning algorithms to monitor and predict parking space occupancy while using blockchain to ensure data integrity and security. Users can access parking information through a mobile app, which provides real-time updates and secure payment options. The integration of blockchain enhances trust and transparency in the system, making it more reliable for users.

In 2016, Ravi Patel et al. developed a real-time parking management system that uses support vector machines (SVM) to classify parking space availability based on sensor data. The system integrates with a mobile application to provide users with up-to-date information on available parking spots. The machine learning model is trained on historical data to predict future parking trends, aiming to improve parking space utilization and reduce traffic congestion.

In 2017, Anna Green et al. introduced a smart parking system leveraging machine learning and cloud computing. The system utilizes wireless sensors to monitor parking lot occupancy and uploads the data to a cloud platform for analysis. A machine learning algorithm processes this data

to provide real-time and predictive parking information to users via a mobile application. The cloud-based approach ensures scalability and robustness of the system.

In 2018, Mohammed Alkhawaja et al. proposed an intelligent parking guidance system that combines machine learning with vehicle-to-infrastructure (V2I) communication. The system uses machine learning models to analyze parking availability and traffic flow data, which is then communicated to vehicles in real-time. Drivers receive parking spot recommendations and navigation assistance through their in-car systems, reducing the time spent searching for parking.

In 2019, Isabella Brown et al. developed an automated parking system utilizing a hybrid approach of machine learning and heuristic algorithms. The system uses cameras and image processing techniques to detect parking space occupancy. Machine learning models are applied to enhance the accuracy of detection and to predict future parking availability. The mobile application connected to the system provides real-time updates and reservation capabilities for parking spots.

In 2020, Li Zhang et al. designed a parking space detection system using deep learning and Internet of Things (IoT) technologies. The system employs convolutional neural networks (CNNs) to analyse video feeds from parking lot cameras. IoT devices collect additional data on parking space usage, which is processed by machine learning algorithms to provide real-time parking availability information. Users can access this information through a dedicated mobile app.

Chapter 3

REQUIREMENT SPECIFICATION

3.1 Functional Requirements

3.1.1 Vehicle Detection and Guidance:

- The system should detect vehicles entering and exiting the parking area.

3.1.2 Parking Spot Detection:

- The system should accurately detect the occupancy status of parking spots in real-time.
- It should differentiate between occupied and available parking spots.

3.1.3 Space Allocation Optimization:

- The system should optimize parking space allocation to maximize utilization.
- It should consider factors such as vehicle size, proximity to entry points, and user preferences.

3.1.4 User Interface:

- The system should provide a user-friendly interface for drivers to access parking information.
- It should display real-time updates on available parking spots and guide users to these spots.

3.1.5 Predictive Analysis:

- The system should predict future parking availability based on historical data and current trends.
- It should anticipate peak times and adjust parking allocations accordingly.

3.2 Non-Functional Requirements

3.2.1 Accuracy:

- The system should have a high accuracy in detecting available parking spots and guiding vehicles.
- The accuracy of the machine learning models should be regularly evaluated and improved.

3.2.2 Real-time Performance:

- The system should operate in real-time, providing immediate updates to users.
- Response times for detecting available parking spots and guiding vehicles should be minimal.

3.2.3 Scalability:

- The system should be scalable to accommodate varying parking lot sizes and traffic volumes.
- It should handle increasing data loads and user requests without compromising performance.

3.2.4 Reliability:

- The system should be reliable, with minimal downtime and disruptions.
- It should have mechanisms for fault tolerance and recovery in case of failures.

3.2.5 Security:

- The system should ensure the security and privacy of user data.
- It should implement authentication and authorization mechanisms to prevent unauthorized access.

3.3 Software Requirement

- Python,
- OpenCV,
- PyTesseract,
- Pillow,
- Numpy,
- Tesseract OCR Engine

3.4 Hardware Requirement

- Computer System
- Camera

Chapter 4

SYSTEM DESIGN

4.1 Introduction to system design

System design is a crucial phase in the development of an automated parking system using machine learning. It involves creating a detailed blueprint that outlines how various components of the system will interact to achieve the desired functionalities. This phase sets the foundation for the entire project, ensuring that all requirements are met and that the system operates efficiently and reliably.

The primary objective of system design is to transform the identified requirements into a coherent architecture that can be implemented and deployed. This involves determining the hardware and software components needed, defining data flow and processing mechanisms, and establishing communication protocols between different parts of the system. In the context of an automated parking system, this includes integrating sensors, cameras, edge devices, a central server, and a user interface to create a seamless and efficient parking management solution.

Key considerations during the system design phase include scalability, real-time performance, reliability, and security. Scalability ensures that the system can handle varying sizes of parking lots and traffic volumes. Real-time performance is crucial for providing immediate updates to users and making timely decisions on parking space allocation. Reliability guarantees that the system functions correctly with minimal downtime, while security ensures that user data is protected and access to the system is controlled.

4.2 Problem Statement of the proposed system

Develop an Automated Parking System that accurately predicts available parking spaces in commercial parking areas by utilizing edge detection techniques and video sources. The system should consider necessary parameters to achieve high accuracy in real-time parking space detection and prediction.

4.3 Objective of the application

The objectives of the project are to develop an automated parking system that utilizes real-time data analysis to optimize parking space utilization, identify peak hours, and improve revenue collection. The system aims to enhance the overall user experience by providing easy access to available parking spaces and simplifying parking procedures. Additionally, the project will develop and deploy machine learning algorithms to accurately detect and recognize license plate numbers under varying environmental conditions, ensuring reliable and efficient parking management.

4.4 Methodology of the system

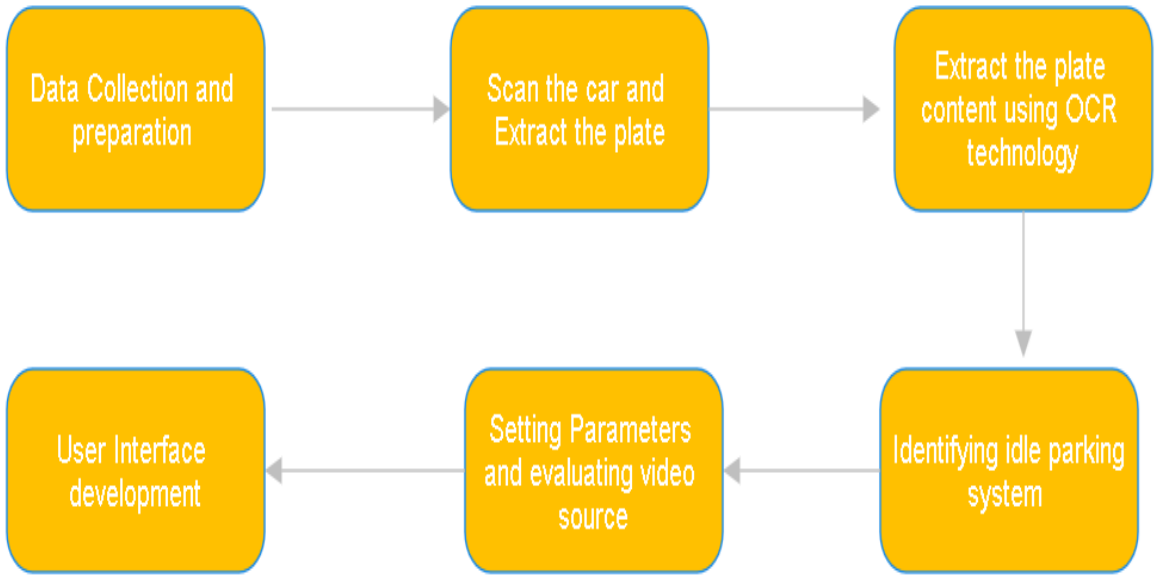


Figure 4.1 Methodology

4.4.1 Data Collection and Preparation

The initial step in developing the automated parking system involves extensive data collection and preparation. High-resolution cameras are strategically installed at entry and exit points of the parking lot to capture images and videos of vehicles. These cameras operate under various

conditions, including different times of day and varying weather scenarios, to ensure a comprehensive dataset. In addition to visual data, occupancy sensors are deployed to log the status of parking spots continuously. This rich dataset includes images, videos, sensor data, and metadata such as timestamps and environmental conditions, providing the necessary information for training machine learning models.

Once the data is collected, it undergoes a thorough preparation process. This includes annotating images and videos to label parking spots and indicate occupancy status using tools like LabelImg or CVAT. Data cleaning is performed to remove noisy, corrupted, or poor-quality samples, ensuring the dataset's accuracy and consistency. Augmentation techniques, such as rotation, scaling, and color adjustments, are applied to increase dataset diversity. The data is then normalized and scaled to maintain uniformity, and split into training, validation, and test sets, ensuring balanced class distributions. Feature extraction and engineering are conducted to derive meaningful inputs for the models, followed by storage in a structured database with backup and versioning mechanisms to ensure data integrity.

4.4.2 Scan the Car and Extract the Plate

The system component responsible for scanning vehicles and extracting license plate information utilizes high-resolution cameras to capture clear images of each vehicle's license plate as it enters the parking area. These images are then pre-processed to enhance quality, involving steps such as noise reduction, contrast enhancement, and conversion to grayscale. Edge detection algorithms, such as Canny edge detection, are applied to identify the contours of the license plate. Morphological operations are used to highlight the rectangular shape of the plate, and contour detection methods localize the region of interest (ROI) containing the license plate.

After localizing the license plate, the system segments it into individual characters using connected component analysis, ensuring each character is isolated and ready for recognition. This segmentation process is crucial for accurate character recognition, as it separates overlapping or closely positioned characters. By integrating these advanced image processing techniques, the system effectively identifies and extracts the license plate from the captured images, setting the stage for accurate optical character recognition (OCR).

4.4.3 Extract the Plate Content Using OCR Technology

With the license plate region and characters isolated, the next step involves using optical character recognition (OCR) technology to extract the text content from the license plate. PyTesseract, a Python wrapper for the Tesseract OCR engine, is utilized to perform this task. Preprocessing steps specific to OCR, such as thresholding and binarization, are applied to the segmented characters to enhance readability and recognition accuracy. PyTesseract then processes these pre-processed images to recognize and extract the alphanumeric characters from the license plate.

Post-processing techniques are employed to correct common OCR errors and validate the extracted text. This may include spelling correction algorithms and cross-referencing the recognized text with a database of known license plate formats to ensure consistency and accuracy. By leveraging OCR technology, the system converts the visual information of the license plate into machine-readable text, which can be used for various purposes such as tracking vehicle entries and exits, billing, and monitoring parking usage.

4.4.4 User Interface Development

Developing a user interface (UI) is a crucial aspect of the automated parking system, as it facilitates interaction between the system and its users. The UI is designed to provide real-time updates on available parking spaces, guiding users to vacant spots efficiently. It includes a dashboard that displays parking lot occupancy, entry and exit times, and license plate information. The UI must be intuitive and user-friendly, ensuring that users can easily access and interpret the information provided.

The development process involves creating wireframes and prototypes to visualize the UI design. Feedback from potential users is incorporated to refine the design, ensuring it meets user needs and expectations. The UI is then developed using web development technologies such as HTML, CSS, and JavaScript, or mobile app frameworks like React Native or Flutter. Integrating the UI with the backend system ensures that it receives real-time data, enabling users to make informed decisions quickly. This interface not only enhances user experience but also streamlines the overall parking process.

4.4.5 Setting Parameters and Evaluating Video Sources

Setting parameters and evaluating video sources are essential steps to ensure the accuracy and reliability of the automated parking system. Parameters such as camera resolution, frame rate, and positioning are optimized to capture high-quality images and videos. The evaluation process involves testing different camera setups to determine the best configuration for capturing clear and detailed images of license plates and parking spots under various conditions.

Once the optimal parameters are set, the video sources are continuously monitored and evaluated to maintain their performance. This includes regular checks and adjustments to account for changes in lighting, weather conditions, and other environmental factors that might affect image quality. By carefully setting and evaluating these parameters, the system can reliably detect and recognize vehicles and license plates, ensuring high accuracy in real-time parking management.

4.4.6 Identifying Ideal Parking System

Identifying the ideal parking system involves analysing the data collected from various sensors and cameras to determine the most efficient way to utilize parking spaces. Machine learning algorithms are developed and trained to predict parking availability, identify peak hours, and optimize space utilization. These algorithms consider factors such as historical occupancy data, time of day, and traffic patterns to make accurate predictions and decisions.

The system continuously learns and adapts based on new data, improving its predictions and optimizations over time. This adaptive capability ensures that the parking system remains efficient and effective in different scenarios and conditions. By integrating these advanced data analysis and machine learning techniques, the system not only enhances the user experience by reducing the time spent searching for parking but also increases revenue for parking lot operators by optimizing space utilization and identifying peak usage times.

4.4.7 Automated Ticketing

Automated ticketing systems have revolutionized the management of parking facilities by leveraging advanced technologies to streamline operations and enhance user experience. In our automated parking system, the process begins with the scanning of a vehicle's number plate using

Tesseract, an OCR (Optical Character Recognition) engine. This scanned data is immediately converted to text, capturing the vehicle's registration number. Upon entry, the system records the time the vehicle enters the parking slot, storing this information in a MongoDB database. This seamless integration ensures that all data is accurately logged in real-time, eliminating manual entry errors and enhancing efficiency.

The use of the MERN stack—comprising MongoDB, Express.js, React, and Node.js—enables a robust and dynamic application. MongoDB serves as the backend database, efficiently handling large volumes of data with its flexible schema design. Express.js and Node.js power the server-side operations, ensuring smooth handling of HTTP requests and efficient interaction with the database. React, on the client side, provides an intuitive and responsive interface for users, facilitating ease of use and interaction with the system.

When the vehicle is scanned again upon exit, the system records the current time and retrieves the corresponding entry time from the database. It then calculates the total parking duration and automatically generates a bill based on predefined rates. This bill is printed instantly, providing a quick and hassle-free checkout process for users. The entire process, from entry to exit, is fully automated, significantly reducing the need for manual intervention and speeding up the flow of vehicles in and out of the parking facility.

This automated ticketing system not only improves operational efficiency but also enhances security and accuracy. By automating data capture and processing, the system minimizes human error and ensures that all transactions are accurately documented. Moreover, the seamless integration of modern technologies in our MERN stack application offers a scalable and reliable solution for modern parking management needs.

Chapter 5

RESULTS

Snapshots

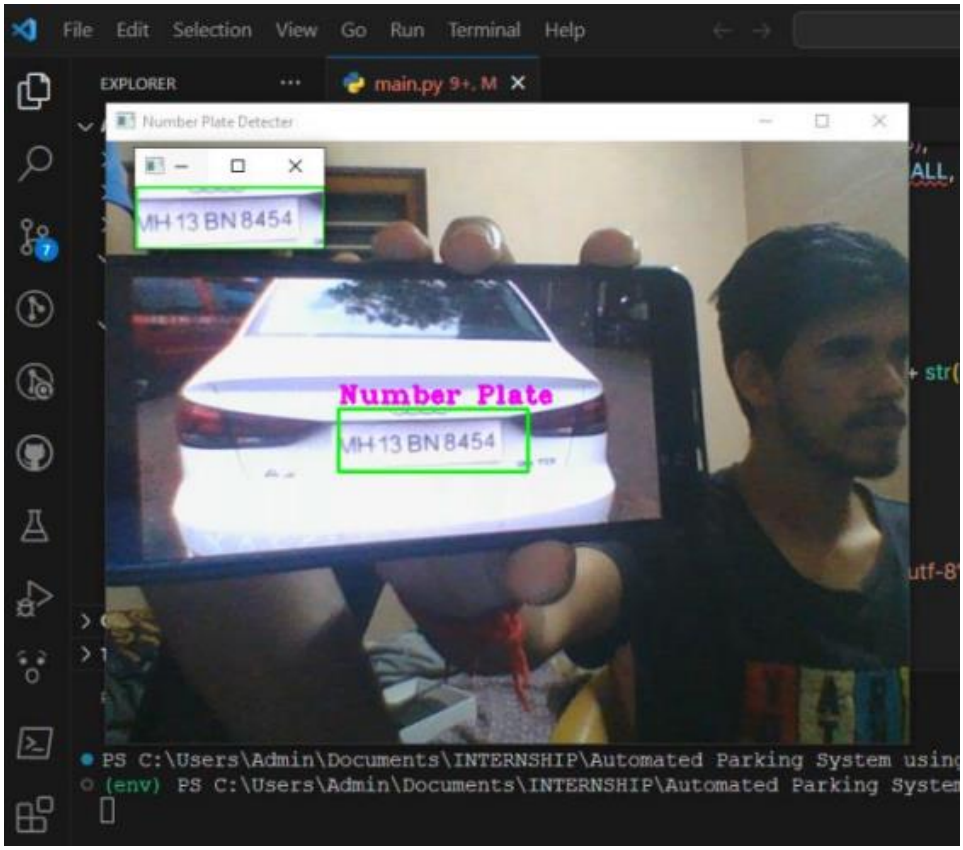


Figure 5.1: Scanning the Car for Number Plate

Initially the car is scanned to detect the number plate. The program window in the foreground shows the detected text highlighted within a bounding box, indicating that the software correctly identified the number plate from a live or saved image.

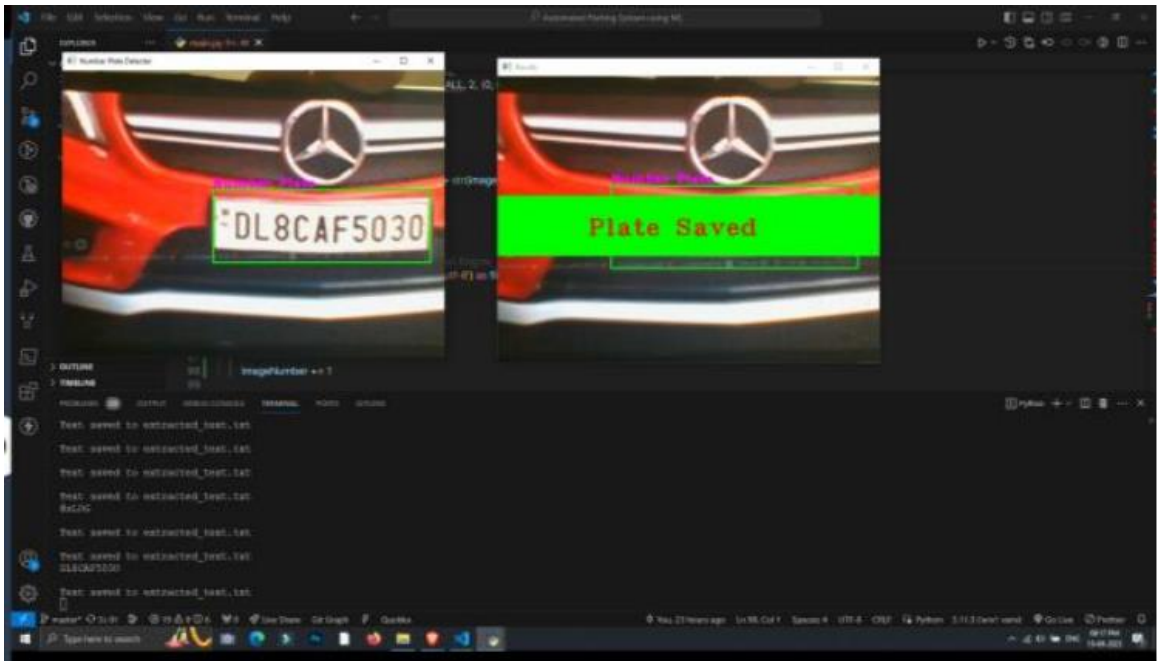


Figure 5.2: Extract plate content using OCR Technology

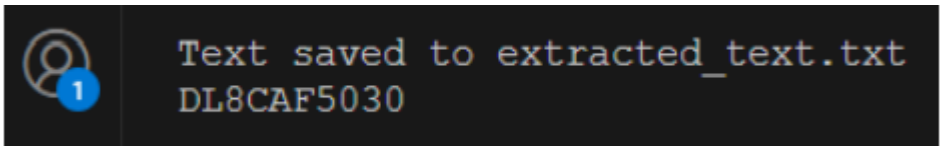


Figure 5.3: Saving the extracted plate content

The text content inside the plate is extracted using OCR Technology and is saved in extracted_text.txt file.

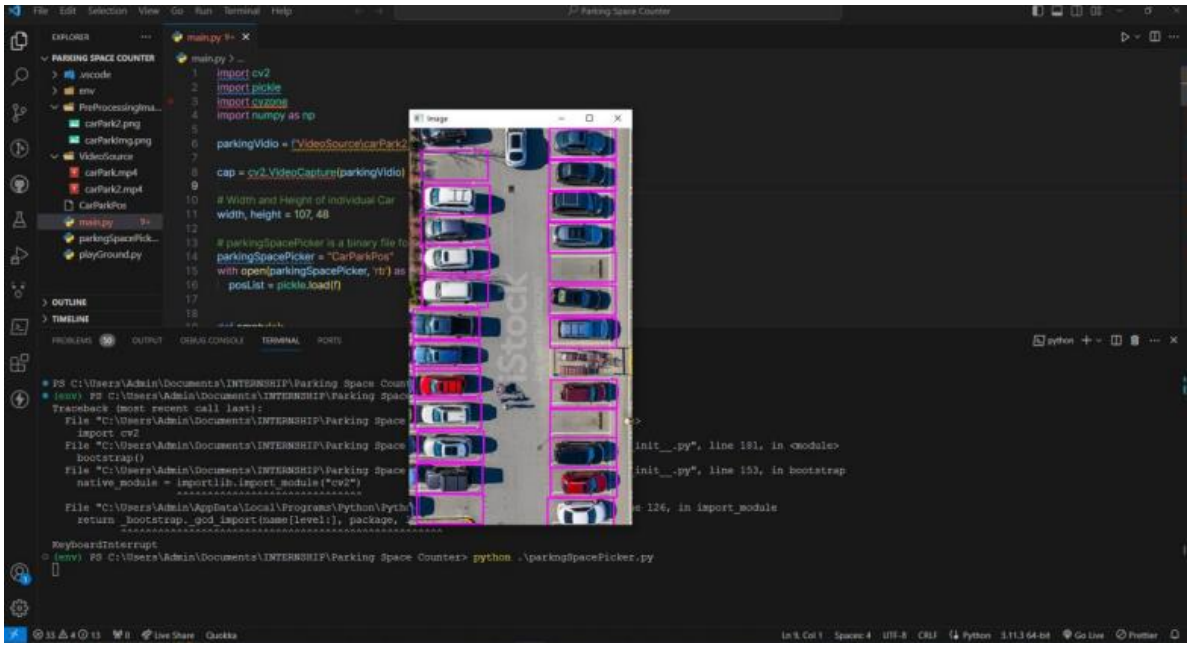


Figure 5.4: Picking the parking Space

Initially, each individual parking space is selected by the administrator to identify all the available parking spaces. This is done only once.

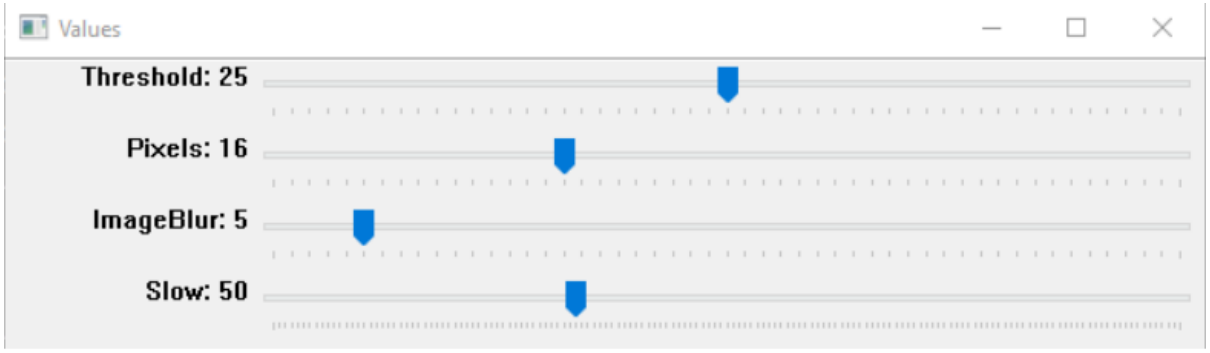


Figure 5.5: Setting the parameters

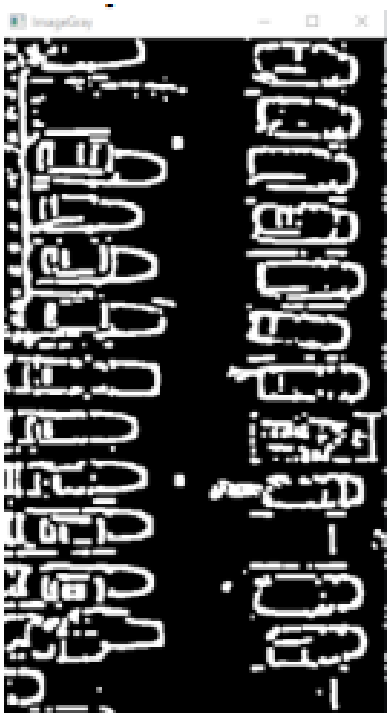


Figure:5.6: Grey scaling the video source

The parameters are set based on our requirements to get an accurate prediction from the grayscale version of the video source. The number of pixels in a particular box is calculated to determine whether the parking space is free or not.



Figure 5.7: Free space available

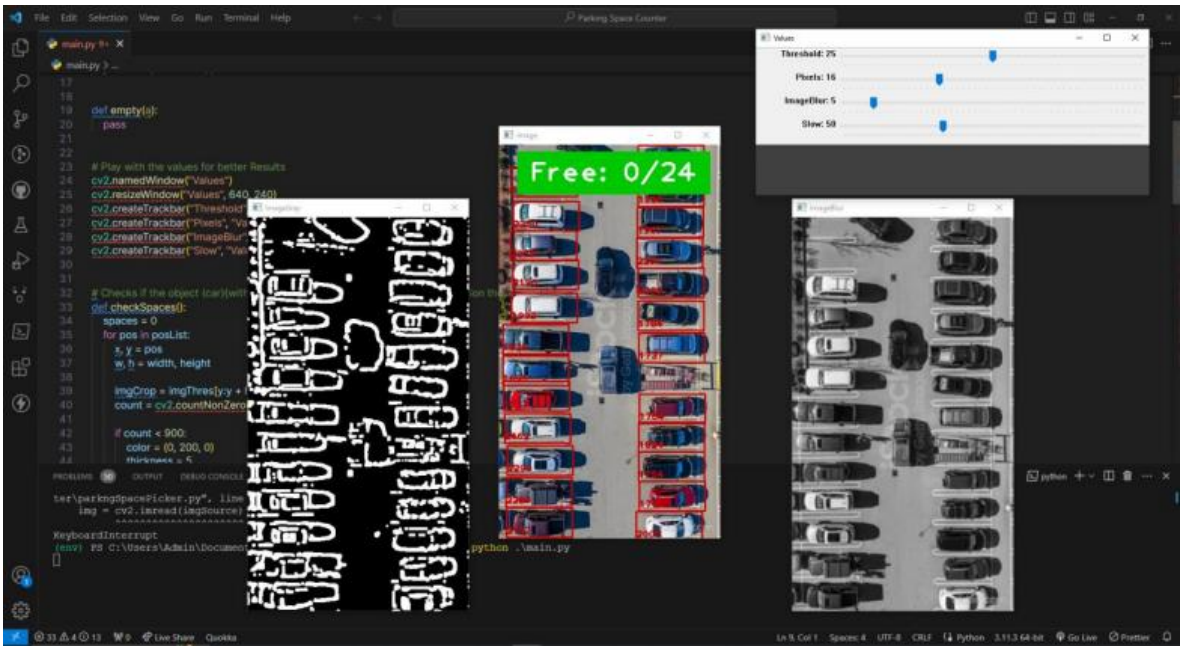


Figure 5.8: Overall View of Automated Parking System



Figure 5.9: Interface for uploading the number plate

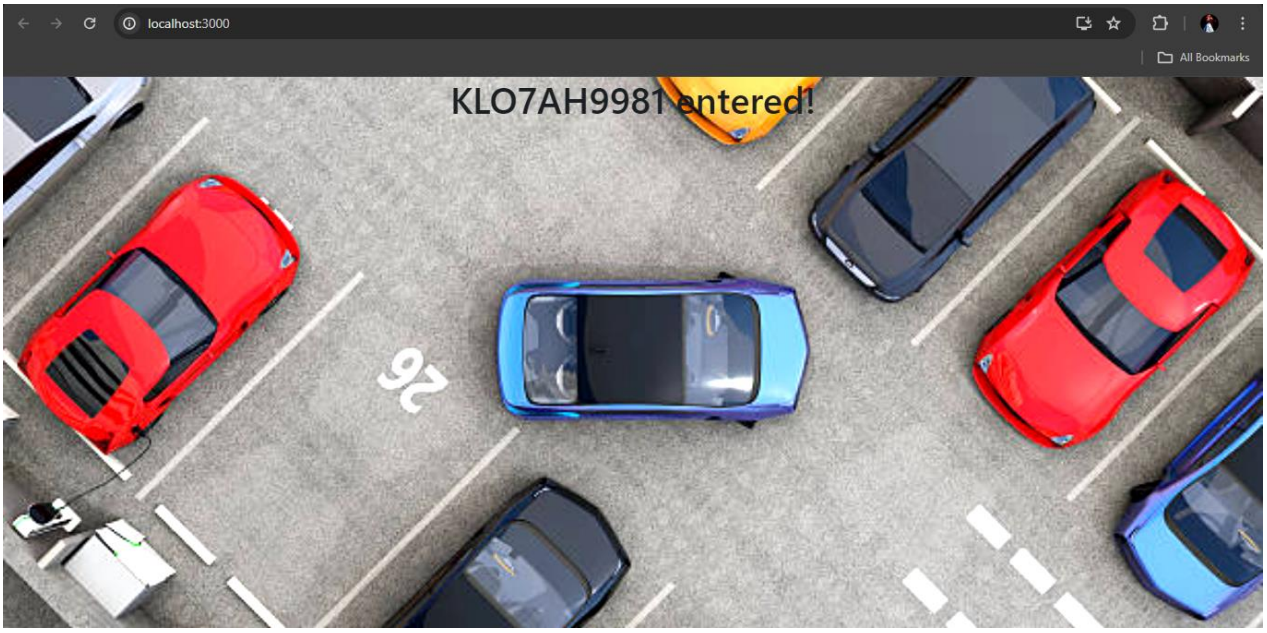


Figure 5.10: Vehicle entered the parking lot

When a vehicle number plate is scanned, we upload the number plate image, as shown in Figure 5.9. If the number plate does not exist in the database, that means the vehicle is entering the parking lot. Hence, the number plate as well as the time the vehicle entered are stored in the database. If the number plate already exists in the database, then we check whether its entered field is true or not. If it is false, then it indicates that the vehicle is entering the parking lot, so we update the time entered field of the vehicle.

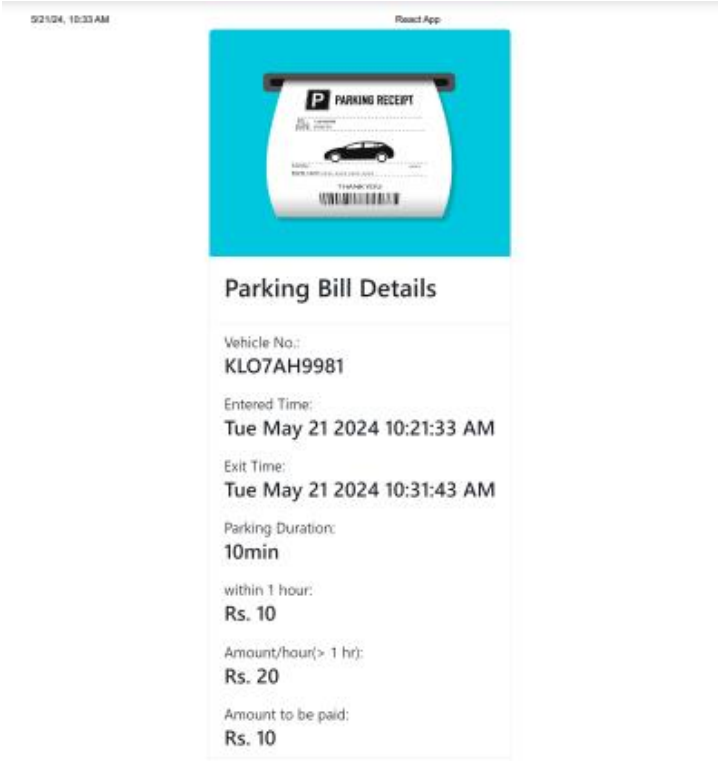


Figure 5.11: Ticket generated

If the entered field of the vehicle record is true, then a ticket is generated based on the logic that if the vehicle was parked for a duration of less than an hour, then the amount payable would be Rs. 10, else they would have to pay Rs. 20 for each hour spent. This can be changed based on the administrator's requirements.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion

Parking management in cities is facing serious challenges due to rapid urban growth and the increasing number of cars. Traditional parking systems, which rely on manual processes, are not able to handle the demand efficiently. This leads to traffic congestion, poor use of parking spaces, and more pollution. However, automated parking systems offer a promising solution to these problems.

Automated parking systems use advanced technologies like sensors, cameras, and machine learning algorithms to improve how parking is managed. They can detect open parking spots, guide drivers to these spots, and optimize space usage in real-time. This makes parking more efficient and convenient, reducing traffic jams and pollution. Additionally, machine learning helps these systems continuously learn and improve, making them more effective over time.

Adopting automated parking systems not only solves immediate parking issues but also supports broader goals like sustainability and smart urban development. By making better use of parking spaces and reducing congestion, these systems help create cleaner and more efficient cities. As cities grow, using automated parking systems is an important step towards building smarter and more sustainable urban environments. Embracing these technologies will make parking less of a hassle and more integrated into the flow of urban life.

7.2 Future Enhancement

1. **Integration with Smart City Infrastructure:** Enhance the automated parking system by integrating it with broader smart city initiatives. This could involve connecting parking data with other urban systems such as traffic management, public transportation, and environmental monitoring to create a more holistic approach to urban mobility and sustainability.
2. **Enhanced User Experience:** Continuously improve the user interface of the automated parking system to provide a more intuitive and personalized experience for drivers. This

could include features such as mobile apps for booking parking in advance, real-time navigation assistance within parking facilities, and seamless integration with payment systems.

3. **Predictive Maintenance:** Implement predictive maintenance algorithms to anticipate and prevent potential failures in the automated parking system's hardware components. By analysing sensor data and identifying patterns indicative of impending issues, proactive maintenance can be conducted to minimize downtime and ensure uninterrupted operation.
4. **Dynamic Pricing and Revenue Optimization:** Introduce dynamic pricing mechanisms based on real-time demand and supply dynamics to optimize revenue generation for parking operators. By adjusting parking fees dynamically according to factors such as occupancy levels, time of day, and special events, the system can maximize revenue while incentivizing efficient space utilization.
5. **Expanded Integration with Transportation Modes:** Explore opportunities to integrate the automated parking system with other transportation modes such as ride-sharing services, electric vehicle charging stations, and bike-sharing programs. By providing seamless connectivity between different modes of transportation, the system can encourage multimodal commuting and reduce reliance on private car ownership.

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